

Notes: Bajari, Benkard, and Levin (Econometrica, 2007)

Estimating Dynamic Models of Imperfect Competition

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1 Big picture

- **Question.** How can applied researchers estimate dynamic games of imperfect competition when computing a Markov perfect equilibrium repeatedly is computationally infeasible and when the model may have multiple equilibria?
- **Core answer.** The paper proposes a two-step estimator. First, estimate the policy functions and state transitions directly from observed behavior. Second, recover structural parameters by imposing the optimality inequalities implied by Markov perfect equilibrium.
- **Key idea.** In equilibrium, observed policies reveal firms' beliefs about rivals and the state transition. Once these beliefs are recovered from data, structural parameters can be estimated without solving the equilibrium of the game even once.
- **Why this matters.** Dynamic oligopoly models such as Ericson-Pakes and Pakes-McGuire are central for studying entry, exit, investment, product quality, advertising, R&D, and capacity. But standard nested fixed-point estimation is difficult because every candidate parameter vector requires computing equilibrium.
- **Estimator class.** The authors develop simulated minimum distance estimators based on equilibrium inequalities. The inequalities require observed equilibrium behavior to yield at least as much value as any feasible deviation.
- **Discrete and continuous actions.** The framework covers dynamic discrete choice, dynamic oligopoly, entry/exit, investment, pricing, and other continuous controls. This is broader than contemporaneous Hotz-Miller-style estimators designed mainly for discrete choices.
- **Multiplicity advantage.** Multiple equilibria are less damaging because the first stage estimates the actual equilibrium strategies observed in the data. The second stage asks which structural parameters rationalize this observed equilibrium.
- **Computational advantage.** Forward simulation replaces repeated equilibrium computation. If period profits are linear in the structural parameters, simulated values are also linear in the parameters, reducing computation substantially.
- **Identification nuance.** The second stage can identify a point or a set. If the optimality inequalities do not point identify the parameters, the paper gives a bounds estimation approach.
- **Main lesson.** Dynamic games can be estimated by using observed play as a revealed equilibrium, then searching for parameters under which that play is approximately optimal.

2 Incremental contribution of the paper

2.1 Contribution relative to nested fixed-point structural estimation

- Earlier structural estimation methods, following Rust, typically require solving the dynamic programming problem or equilibrium model inside an outer parameter search.
- In dynamic games, this becomes especially costly because each candidate parameter vector may require solving a full Markov perfect equilibrium.
- Bajari, Benkard, and Levin make estimation feasible by avoiding equilibrium computation altogether.

Interpretation: The key incremental contribution is a practical estimation strategy for dynamic strategic environments where equilibrium computation is the binding constraint.

2.2 Contribution relative to Hotz–Miller two-step estimation

- Hotz and Miller showed that conditional choice probabilities can be used to recover value differences in single-agent dynamic discrete choice models.
- This paper extends the two-step logic to dynamic games of imperfect competition.
- It also avoids being limited to finite discrete actions by using monotone-choice arguments and forward simulation.

Interpretation: The paper turns the two-step single-agent insight into a broad method for dynamic oligopoly, including continuous controls such as investment, price, capacity, or advertising.

2.3 Contribution relative to contemporaneous dynamic game estimators

- Pakes, Ostrovsky, and Berry; Pesendorfer and Schmidt-Dengler; and Aguirregabiria and Mira develop related two-step estimators for dynamic games, mainly in discrete choice frameworks.
- Bajari, Benkard, and Levin emphasize an inequality-based second stage that is convenient for both discrete and continuous actions.
- The method applies even when exact first-order conditions are difficult, as long as profitable deviations can be simulated.

Interpretation: This paper’s incremental value is generality: it provides a unified simulation-based framework for many dynamic industrial organization models.

2.4 Contribution relative to partial identification and incomplete models

- The paper explicitly recognizes that equilibrium inequalities may not point identify all parameters.
- Rather than forcing point estimates, it develops a bounds approach for the parameter set consistent with equilibrium optimality.
- This anticipates a broader literature on partial identification and moment inequality models.

Interpretation: The estimator is honest about what the data and equilibrium restrictions can identify. This is particularly important in dynamic games, where multiple equilibria and limited variation can weaken identification.

3 Model primitives

3.1 Dynamic game environment

There are N firms. At each time t , firm i observes a public state s_t and a private shock ν_{it} , then chooses action $a_{it} \in A_i$. The action profile is

$$a_t = (a_{1t}, \dots, a_{Nt}). \quad (1)$$

Firm i ’s one-period profit is

$$\pi_i(a_t, s_t, \nu_{it}; \theta), \quad (2)$$

where θ is the vector of structural parameters to estimate. The state evolves according to

$$s_{t+1} \sim P(\cdot \mid a_t, s_t). \quad (3)$$

Interpretation: Actions can affect future market structure, future productivity, future product quality, or future demand conditions. This is what makes the model dynamic.

3.2 Markov strategies and Markov perfect equilibrium

A Markov strategy for firm i is

$$\sigma_i : S \times V_i \rightarrow A_i. \quad (4)$$

A strategy profile $\sigma = (\sigma_1, \dots, \sigma_N)$ induces the value function

$$V_i(s; \sigma) = \mathbb{E}_\nu \left[\pi_i(\sigma(s, \nu), s, \nu_i) + \beta \int V_i(s'; \sigma) dP(s' | \sigma(s, \nu), s) \right]. \quad (5)$$

The profile σ is a Markov perfect equilibrium if for every firm i , state s , and alternative Markov strategy σ'_i ,

$$V_i(s; \sigma) \geq V_i(s; \sigma'_i, \sigma_{-i}). \quad (6)$$

Interpretation: The entire estimator is built around this inequality. The observed policy should be at least as valuable as any deviation when rivals follow their observed policies.

3.3 Example 1: Dynamic discrete choice

In the dynamic discrete choice example, a single agent chooses whether to maintain or replace a machine. The state is machine age s , and the action is $a \in \{0, 1\}$. Profits are

$$\pi(a, s, \nu; \theta) = \begin{cases} -\mu s + \nu(0), & a = 0, \\ -R + \nu(1), & a = 1. \end{cases} \quad (7)$$

Interpretation: This example connects the method to the classic Rust bus-engine replacement model. It provides a simple environment to verify that the estimator can recover dynamic cost parameters.

3.4 Example 2: Dynamic oligopoly

In the dynamic oligopoly example, incumbents have states $z_i \in \{1, \dots, \bar{z}\}$ representing product quality, productivity, or capacity. Firms choose investment $I_i \geq 0$ and exit decisions. Incumbent profits are

$$q_i(p_i - mc(q_i, s; \mu)) - C(I_i, \nu_i; \xi). \quad (8)$$

Potential entrants observe entry costs and choose whether to enter. The model is similar to Pakes–McGuire and Ericson–Pakes.

Interpretation: This example is the main application target: dynamic industry competition with continuous investment, entry, exit, and forward-looking strategic interaction.

4 First-stage estimation

4.1 Data-generating process

The data are assumed to be generated by one Markov perfect equilibrium. The econometrician observes states and actions across markets or time periods:

$$\{s_t, a_t\}_{t=1}^T. \quad (9)$$

The first-stage objects are:

- the policy functions $\sigma_i(s, \nu_i)$;
- the distribution of private shocks, when needed;
- the law of motion $P(s' | a, s)$.

Interpretation: The first stage reconstructs the equilibrium beliefs that firms have about rivals and state transitions.

4.2 Discrete choice policy functions

For discrete choices, the econometrician estimates conditional choice probabilities

$$\Pr(a_i \mid s). \quad (10)$$

With logit shocks, choice-specific value differences satisfy

$$v_i(a'_i, s) - v_i(a_i, s) = \log \Pr(a'_i \mid s) - \log \Pr(a_i \mid s). \quad (11)$$

Interpretation: This is the Hotz–Miller inversion logic. It recovers value differences without solving the dynamic programming problem.

4.3 Continuous choice policy functions

For continuous actions, the paper assumes a monotone choice condition. If the policy is increasing in private shock ν_i , then

$$F_i(a_i \mid s) = G_i(\sigma_i^{-1}(a_i; s) \mid s; \theta). \quad (12)$$

Therefore,

$$\sigma_i(s, \nu_i) = F_i^{-1}(G_i(\nu_i \mid s; \theta) \mid s). \quad (13)$$

Interpretation: This is the key step that makes the estimator apply to continuous choices. The distribution of observed actions identifies the policy function through monotonicity.

4.4 Estimating state transitions

The transition kernel is estimated from observed state transitions:

$$\hat{P}(s' \mid a, s). \quad (14)$$

In practice, this may involve parametric models, nonparametric frequency estimates, or smoothed estimators when the state space is large.

Interpretation: Transition estimation recovers how current actions shape future states, which is essential for valuing dynamic deviations.

5 Value function estimation by forward simulation

5.1 Forward simulation algorithm

Given an estimated policy profile $\hat{\sigma}$ and transition rule $\hat{P}$, the value of a strategy profile is estimated by simulation. Starting from state $s_0 = s$:

1. Draw private shocks ν_{it} for every firm.
2. Compute actions $a_{it} = \sigma_i(s_t, \nu_{it})$.
3. Compute period profits $\pi_i(a_t, s_t, \nu_{it}; \theta)$.
4. Draw the next state s_{t+1} from $\hat{P}(\cdot \mid a_t, s_t)$.
5. Repeat and discount the simulated profit stream.

The simulated value is

$$\hat{V}_i(s; \sigma; \theta) = \frac{1}{R} \sum_{r=1}^R \sum_{t=0}^T \beta^t \pi_i(a_t^r, s_t^r, \nu_{it}^r; \theta). \quad (15)$$

Interpretation: Forward simulation estimates value functions without solving Bellman equations or computing equilibria.

5.2 Linearity simplification

If period profits are linear in structural parameters,

$$\pi_i(a, s, \nu_i; \theta) = \psi_i(a, s, \nu_i)' \theta, \quad (16)$$

then simulated values are also linear:

$$\widehat{V}_i(s; \sigma; \theta) = \widehat{W}_i(s; \sigma)' \theta. \quad (17)$$

Interpretation: This feature makes the second stage computationally simple. Simulated sufficient statistics can be computed once and reused for all candidate parameter values.

6 Second-stage estimation

6.1 Equilibrium inequalities

For each firm i , state s , and deviation strategy σ'_i , equilibrium implies

$$V_i(s; \widehat{\sigma}_i, \widehat{\sigma}_{-i}; \theta) \geq V_i(s; \sigma'_i, \widehat{\sigma}_{-i}; \theta). \quad (18)$$

Define the difference

$$g_{i,s,\sigma'_i}(\theta) = V_i(s; \widehat{\sigma}; \theta) - V_i(s; \sigma'_i, \widehat{\sigma}_{-i}; \theta). \quad (19)$$

Then the model requires

$$g_{i,s,\sigma'_i}(\theta) \geq 0. \quad (20)$$

Interpretation: Parameters are chosen so that no tested deviation is profitable.

6.2 Simulated minimum distance estimator

The estimator minimizes violations of the equilibrium inequalities:

$$\widehat{\theta} = \arg \min_{\theta \in \Theta} \sum_{i,s,\sigma'_i} [\min\{g_{i,s,\sigma'_i}(\theta), 0\}]^2. \quad (21)$$

Interpretation: If all inequalities hold exactly, the objective is zero. With sampling error and simulation error, the estimator chooses parameters that make observed behavior as close as possible to equilibrium best response behavior.

6.3 Choice of deviations

The researcher need not test every possible deviation. Useful deviations include:

- changing investment by a small amount;
- changing entry or exit probabilities;
- choosing observed actions from other states;
- parametric perturbations of the estimated policy function;
- deviations that target specific parameters, such as entry cost or scrap value.

Interpretation: The power of the estimator depends on the set of deviations. If deviations do not move a particular parameter’s payoff-relevant margin, that parameter may be weakly identified.

6.4 Bounds estimation

When inequalities do not point identify θ , the identified set is

$$\Theta_I = \{\theta \in \Theta : g_{i,s,\sigma'_i}(\theta) \geq 0 \text{ for all tested } (i, s, \sigma'_i)\}. \quad (22)$$

The paper adapts bounds estimation to dynamic games by characterizing the set of parameters that rationalize observed policies.

Interpretation: This is important because dynamic games often contain limited variation and many parameters. Partial identification is not a failure; it is a truthful description of what the model and data imply.

7 Serially correlated unobserved state variables

- The baseline model assumes that the econometrician observes the payoff-relevant state variables, apart from private shocks.
- The paper discusses extensions with serially correlated unobserved states.
- One approach is iterative: start with first-stage estimates, estimate structural parameters, use the structure to update beliefs about unobserved states, then repeat.
- This resembles iterative versions of the Hotz–Miller estimator and the Aguirregabiria–Mira nested pseudo-likelihood approach.

Interpretation: Serially correlated unobserved heterogeneity is a major challenge because firms condition on information that the econometrician lacks. The paper does not solve this completely but explains how the framework can be extended.

8 Monte Carlo applications

8.1 Dynamic discrete choice Monte Carlo

The first Monte Carlo study estimates the machine replacement model. The structural parameters are μ and R . The exercise evaluates finite-sample bias, real standard errors, subsampling standard errors, and coverage.

Interpretation: The results show that the estimator performs well in a familiar dynamic discrete choice setting. This validates the method in a simpler environment before moving to dynamic oligopoly.

8.2 Dynamic oligopoly Monte Carlo

The second Monte Carlo applies the estimator to a dynamic oligopoly model with investment, entry, and exit. The payoff parameters include investment cost, scrap value, and entry-cost distribution parameters.

Interpretation: This is the main proof of concept for dynamic industrial organization. It demonstrates that the estimator can recover dynamic primitives without computing equilibria during estimation.

8.3 Alternative method-of-moments estimator

The paper also studies a method-of-moments estimator based on aggregated moment conditions. This can reduce finite-sample bias because aggregation averages first-stage errors.

Interpretation: The moment estimator trades off some information for robustness to noisy first-stage estimates.

9 Identification and empirical design

9.1 What identifies structural parameters

- Policy functions reveal how firms respond to states and private shocks.
- State transitions reveal how actions affect future environments.
- Equilibrium inequalities connect observed behavior to dynamic payoff parameters.
- Deviations identify parameters by asking how profits would change if firms altered their actions.

Interpretation: Identification is revealed-preference based: observed equilibrium choices must be better than deviations.

9.2 Strengths

- Avoids repeated equilibrium computation.
- Handles multiple equilibria by conditioning on the equilibrium observed in the data.
- Works for both discrete and continuous controls.
- Uses simulation rather than analytic value-function solutions.
- Allows partial identification through inequalities.

9.3 Limitations

- The first stage must recover policy functions accurately.
- Large state spaces can make first-stage estimation noisy.
- Results can depend on the set of deviations used in the second stage.
- Unobserved serially correlated state variables are difficult.
- Equilibrium is assumed to be Markov perfect and stable across the data-generating environment.

Interpretation: The estimator solves the equilibrium-computation problem, but it does not remove the need for credible state variables, rich data, and well-chosen deviations.

10 Tables and figures

10.1 Tables I and II: Monte Carlo setup and discrete choice performance

Read:

- Table I reports the dynamic discrete choice Monte Carlo estimates for the maintenance cost parameter μ and replacement cost R .
- Estimates are close to unbiased except in very small samples.
- Subsampling standard errors track real standard errors reasonably well.
- Table II reports the dynamic oligopoly parameter values used in the Monte Carlo exercise.
- The parameters govern demand, investment evolution, discounting, investment costs, marginal cost, entry-cost support, and scrap value.

Economic message: Table I validates the estimator in a familiar single-agent environment. Table II defines the richer dynamic oligopoly environment used to stress-test the method.

ESTIMATING DYNAMIC MODELS					
TABLE I					
DYNAMIC DISCRETE CHOICE MONTE CARLO ^a					
	Mean	SE (Real)	5% (Real)	95% (Real)	SE (Subsampling)
$n = 400, n_I = 200, n_s = 1000$					
μ	1.00	0.14	0.79	1.24	0.10
R	4.02	0.53	3.24	4.96	0.39
$n = 200, n_I = 200, n_s = 500$					
μ	0.99	0.18	0.72	1.37	0.17
R	4.00	0.78	2.94	5.95	0.86
$n = 100, n_I = 200, n_s = 250$					
μ	0.94	0.32	0.47	1.48	0.35
R	3.75	1.26	1.92	5.70	1.15
$n = 50, n_I = 200, n_s = 150$					
μ	0.89	0.54	0.11	2.03	0.47
R	3.57	2.35	0.60	8.16	2.27

^a 500 Monte Carlo runs; 25 subsamples of size $n/2$.

Table I: Dynamic discrete choice Monte Carlo

TABLE II			
DYNAMIC OLIGOPOLY MONTE CARLO PARAMETERS			
Parameter	Value	Parameter	Value
Demand		Discount factor β	0.925
γ_1	1.5	Investment cost ξ	1
γ_0	0.1	Marginal cost μ	3
M	5	Entry cost distribution	
y	6	ν^l	7
Investment evolution		ν^h	11
δ	0.7	Scrap value ϕ	6
ρ	1.25		

Table II: Dynamic oligopoly Monte Carlo parameters

10.2 Tables III and IV: dynamic oligopoly estimates

Read:

- Table III reports estimates when the entry-cost distribution is treated nonparametrically.
- Table IV reports estimates when the entry-cost distribution is parameterized.
- Investment cost ξ and scrap value ϕ are estimated precisely even with moderate sample sizes.
- Entry-cost distribution parameters are estimated less precisely, indicating that entry behavior provides limited information about these parameters in the Monte Carlo design.

Economic message: The estimator works in a dynamic strategic setting, but identification differs across parameters depending on how strongly observed actions reveal the corresponding margin.

	Mean	SE (Real)	5% (Real)	95% (Real)	SE (Subsampling)
$n = 400, n_I = 500$					
ξ	1.01	0.05	0.91	1.10	0.03
ϕ	5.38	0.43	4.70	6.06	0.39
$n = 200, n_I = 500$					
ξ	1.01	0.08	0.89	1.14	0.05
ϕ	5.32	0.56	4.45	6.33	0.53
$n = 100, n_I = 300$					
ξ	1.01	0.10	0.84	1.17	0.06
ϕ	5.30	0.72	4.15	6.48	0.72

small sample sizes used. Overall the estimates are surprisingly precise given
Table III: Nonparametric entry-cost distribution

	Mean	SE (Real)	5% (Real)	95% (Real)	SE (Subsampling)
$n = 400, n_I = 500$					
ξ	1.01	0.06	0.92	1.10	0.04
ϕ	5.38	0.42	4.68	6.03	0.41
ν^l	6.21	1.00	4.22	7.38	0.26
ν^h	11.2	0.67	10.2	12.4	0.30
$n = 200, n_I = 500$					
ξ	1.01	0.07	0.89	1.13	0.05
ϕ	5.28	0.66	4.18	6.48	0.53
ν^l	6.20	1.16	3.73	7.69	0.34
ν^h	11.2	0.88	9.99	12.9	0.40
$n = 100, n_I = 300$					
ξ	1.01	0.10	0.84	1.17	0.06
ϕ	5.43	0.81	4.26	6.74	0.75
ν^l	6.38	1.42	3.65	8.43	0.51
ν^h	11.4	1.14	9.70	13.3	0.58

Table IV: Parametric entry-cost distribution

10.3 Figures 1 and 2: recovered entry-cost distributions

Read:

- Figures 1 and 2 compare estimated entry-cost distributions with the true distribution for $n = 400$ and $n = 200$.
- The mean estimate is close to the true distribution, especially in larger samples.
- Confidence bands widen as sample size falls.

Economic message: The figures show that even though entry-cost parameters are harder to estimate than investment cost and scrap value, the recovered distribution is broadly informative in moderate samples.

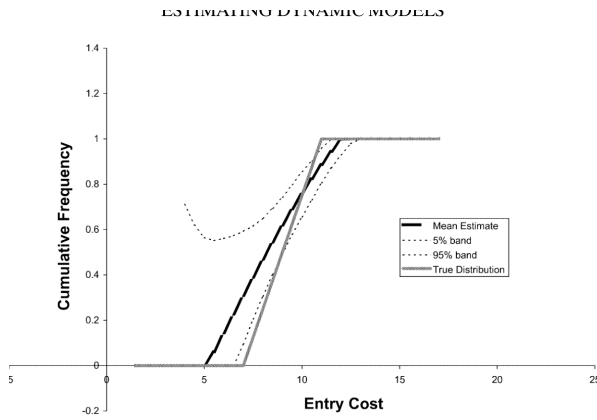


FIGURE 1.—Entry cost distribution for $n = 400$.

Figure 1: Entry cost distribution, $n = 400$

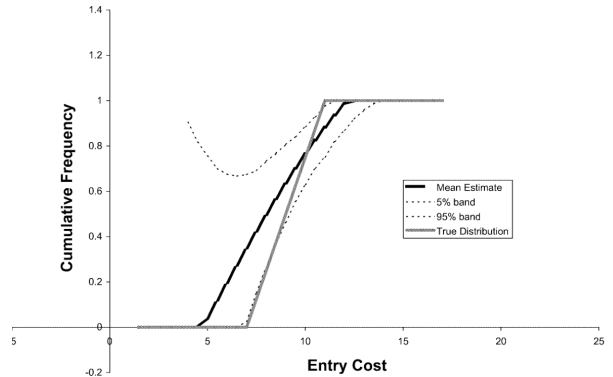


FIGURE 2.—Entry cost distribution for $n = 200$.

Figure 2: Entry cost distribution, $n = 200$

10.4 Figure 3 and Table V: small sample distribution and method-of-moments estimator

Read:

- Figure 3 shows the recovered entry-cost distribution for $n = 100$.
- The mean estimate is more biased and confidence bands are wider in the smaller sample.
- Table V reports the method-of-moments estimator, which reduces some finite-sample bias relative to the inequality estimator.
- The moment estimator performs especially well for the dynamic oligopoly payoff parameters.

Economic message: The comparison illustrates a practical tradeoff. Inequality estimators use detailed optimality restrictions, while aggregated moments can be more robust to noisy first-stage estimates in small samples.

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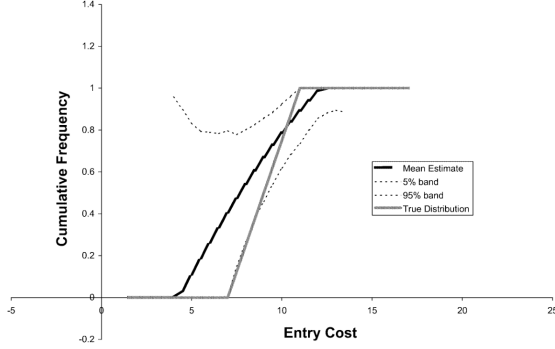


FIGURE 3.—Entry cost distribution for $n = 100$.

Figure 3: Entry cost distribution, $n = 100$

	Mean	SE (Real)	5% (Real)	95% (Real)	SE (Subsampling)
$n = 400, n_I = 500$					
ξ	1.01	0.03	0.96	1.06	0.03
ϕ	6.09	0.18	5.86	6.36	0.05
ν^l	7.04	1.01	5.42	8.35	0.90
ν^h	11.1	0.65	9.97	12.0	0.57
$n = 200, n_I = 500$					
ξ	1.01	0.05	0.92	1.08	0.04
ϕ	6.06	0.24	5.74	6.38	0.10
ν^l	6.89	1.98	3.52	9.16	1.35
ν^h	11.0	1.54	9.21	12.5	0.91
$n = 100, n_I = 300$					
ξ	1.01	0.06	0.90	1.10	0.04
ϕ	6.02	0.34	5.43	6.48	0.15
ν^l	6.86	2.34	2.61	9.16	1.44
ν^h	11.1	1.83	8.73	13.58	1.09

Table V: Method-of-moments estimates

11 Short summary

- The paper proposes a two-step estimator for dynamic games of imperfect competition.
- Step 1 estimates equilibrium policies and state transitions directly from observed data.
- Step 2 estimates structural parameters using equilibrium optimality inequalities.
- The estimator avoids computing Markov perfect equilibria during parameter search.
- It handles both discrete and continuous actions.
- It mitigates the multiple-equilibrium problem by conditioning on observed equilibrium behavior.
- Forward simulation estimates value functions for observed policies and deviations.
- If payoffs are linear in parameters, value functions are linear too, greatly reducing computation.
- The estimator may point identify parameters or characterize an identified set.
- Monte Carlo evidence shows good performance in dynamic discrete choice and dynamic oligopoly examples.

12 Conclusion

12.1 Core conclusion

Bajari, Benkard, and Levin provide a practical way to estimate dynamic oligopoly models without solving the equilibrium at every candidate parameter vector. Their method uses observed equilibrium behavior to recover beliefs and then imposes equilibrium optimality through simulated inequalities.

12.2 Economic intuition

A firm's observed policy reveals what it expects rivals and future states to do. If that policy is part of a Markov perfect equilibrium, it must be at least as valuable as feasible deviations. Structural parameters are therefore those that make the observed policy a best response.

12.3 Takeaway

The paper's central contribution is to turn dynamic game estimation from an equilibrium computation problem into a revealed-preference simulation problem. This greatly expands the set of dynamic competitive models that can be taken to data.